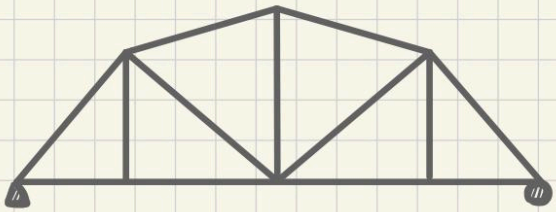
ME 20 Project 1

Ani Gregori Asadourian and Anne Jumper

Part 1: Design and Theoretical Support Reactions

Gambrel Truss



M=8; N=13; R=3

Statically Determinate:

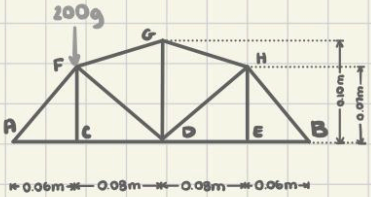
2M = N + R

2(8) = 13+3

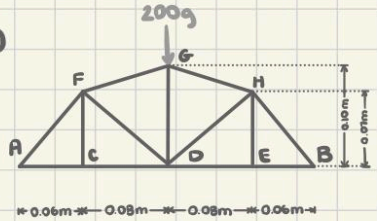
16 = 16 √ yes

Loading conditions

① 200g

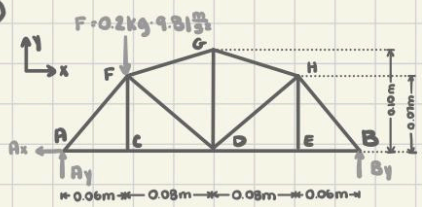


② 200g



Support Reactions

①



Equilibrium Equations:

$\rightarrow \sum F_x = 0$: $A_x = 0$

$+\uparrow \sum F_y = 0$: $A_y + B_y - F = 0$

$A_y + B_y = (0.2\text{kg})(9.81 \frac{m}{s^2})$

$A_y + B_y = 1.962\ N$

$A_y = 1.962\ N - 0.4204\ N$

$A_y = 1.542\ N$

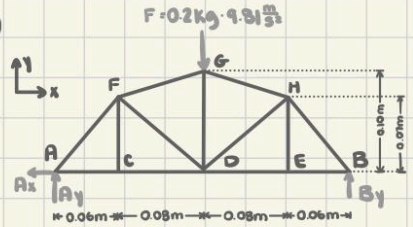
$\circlearrowleft + \sum M_A = 0$: $-F(0.06m) + B_y(0.28m) = 0$

$B_y(0.28m) = (0.2\text{kg})(9.81 \frac{m}{s^2})(0.06m)$

$B_y(0.28m) = 0.11772\ N \cdot m$

$B_y = 0.420\ N$

②



Equilibrium Equations:

$\rightarrow \sum F_x = 0$: $A_x = 0$

$+\uparrow \sum F_y = 0$: $A_y + B_y - F = 0$

$A_y + B_y = (0.2\text{kg})(9.81 \frac{m}{s^2})$

$A_y + B_y = 1.962\ N$

$A_y = 1.962\ N - 0.981\ N$

$A_y = 0.981\ N$

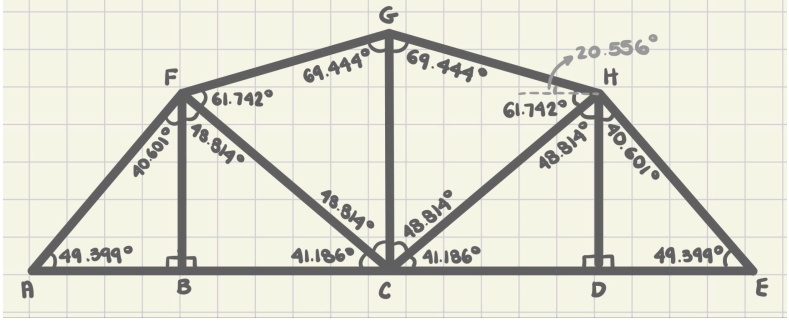
$\circlearrowleft + \sum M_A = 0$: $-F(0.14m) + B_y(0.28m) = 0$

$B_y(0.28m) = (0.2\text{kg})(9.81 \frac{m}{s^2})(0.14m)$

$B_y(0.28m) = 0.27468\ N \cdot m$

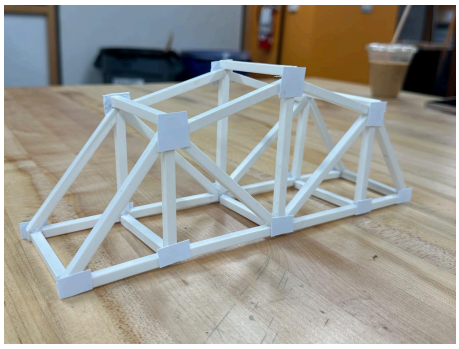
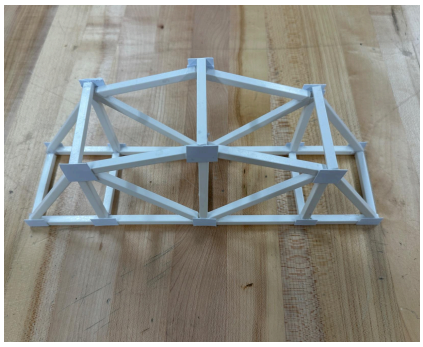
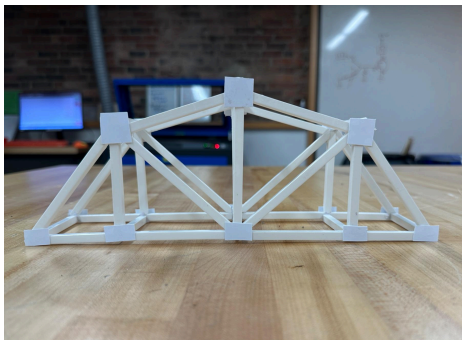
$B_y = 0.981\ N$

Angles Between Members

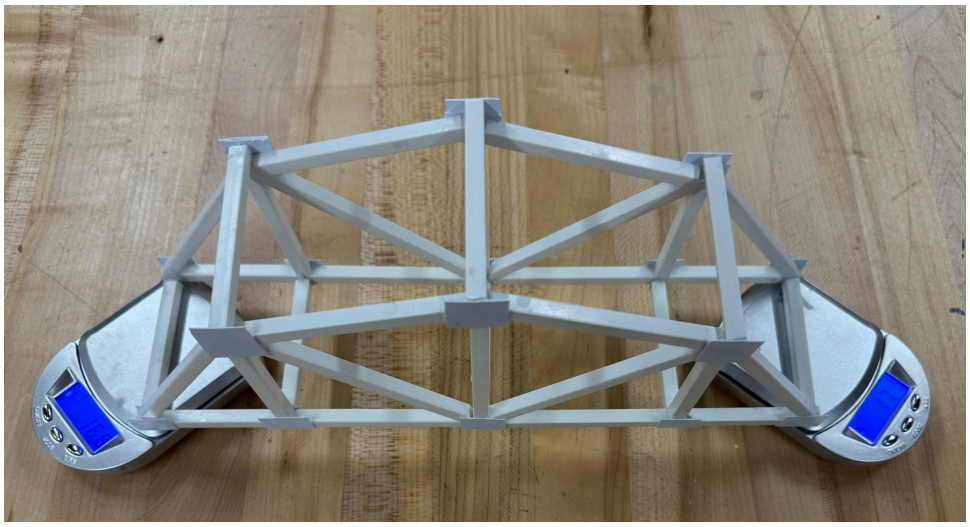


Part 2: Building the Truss and Finding Support Reactions

Photos of Truss

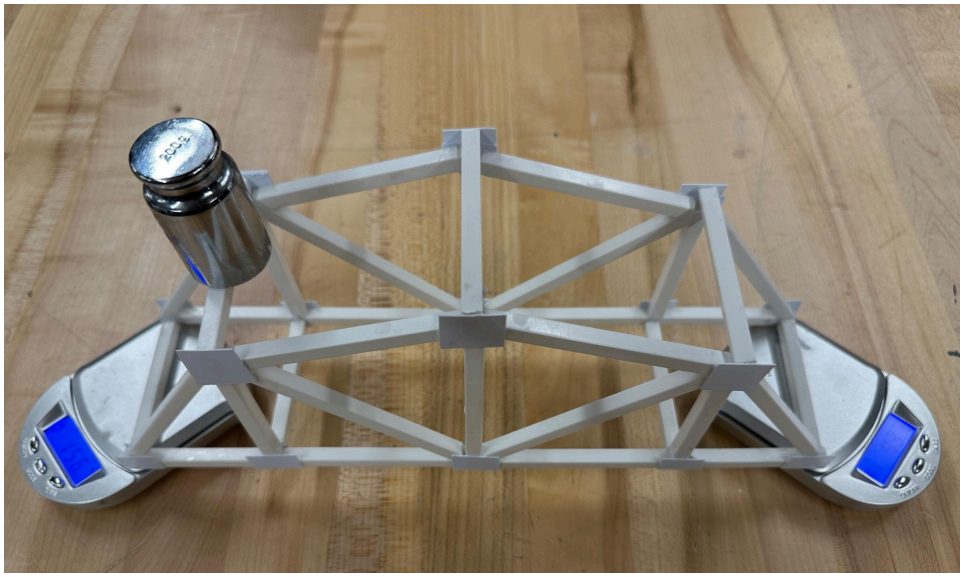




Unloaded Truss (before mass was tared)



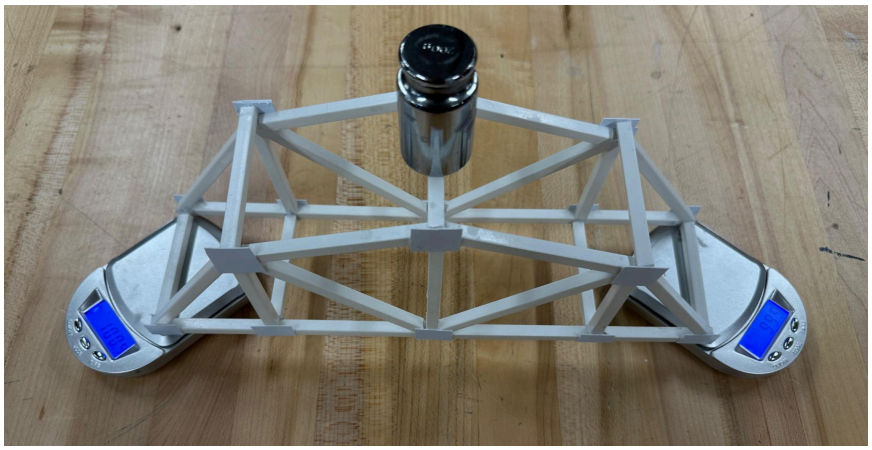
Left Mass: 16.8 grams; Right Mass: 17.1g

Loading Condition 1



Left Mass: 156.7g; Right Mass: 43.7g

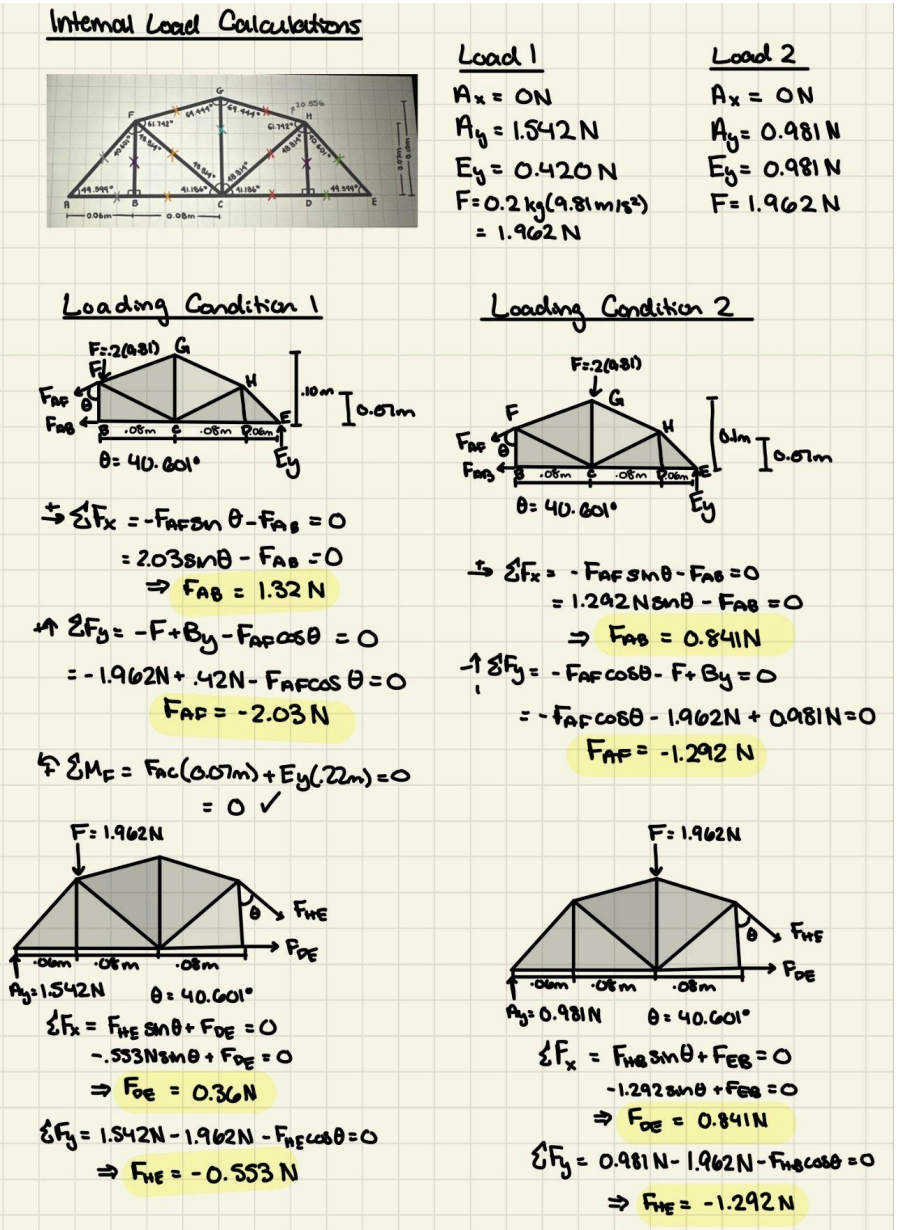
Loading Condition 2



Left Mass: 100.4g; Right Mass: 99.9g

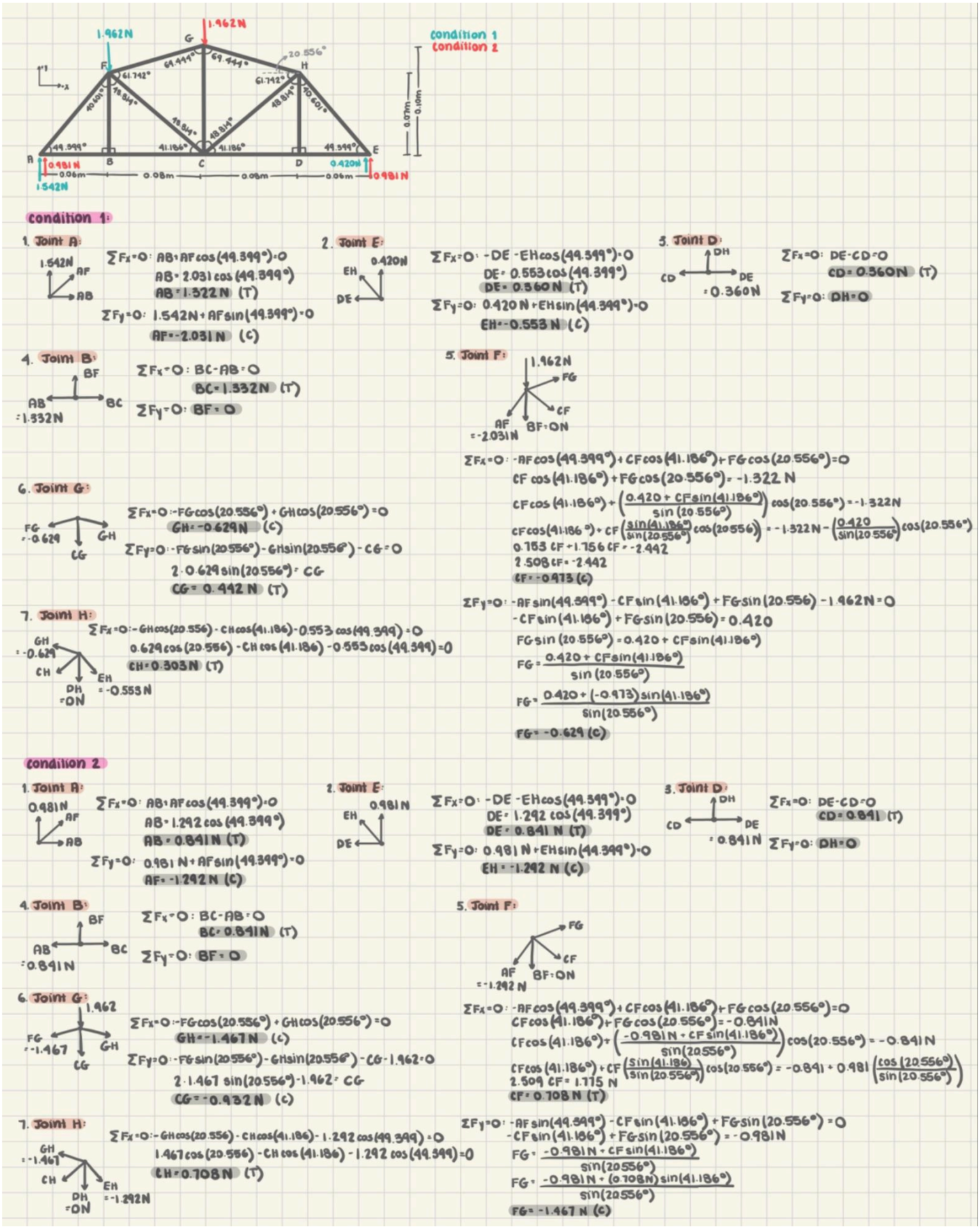
Part 3: Members' Theoretical Internal Loads

Method of Sections:



*Our chosen convention is that tension is positive, thus negative values indicate the member is under compression and positive values indicate the member is under tension

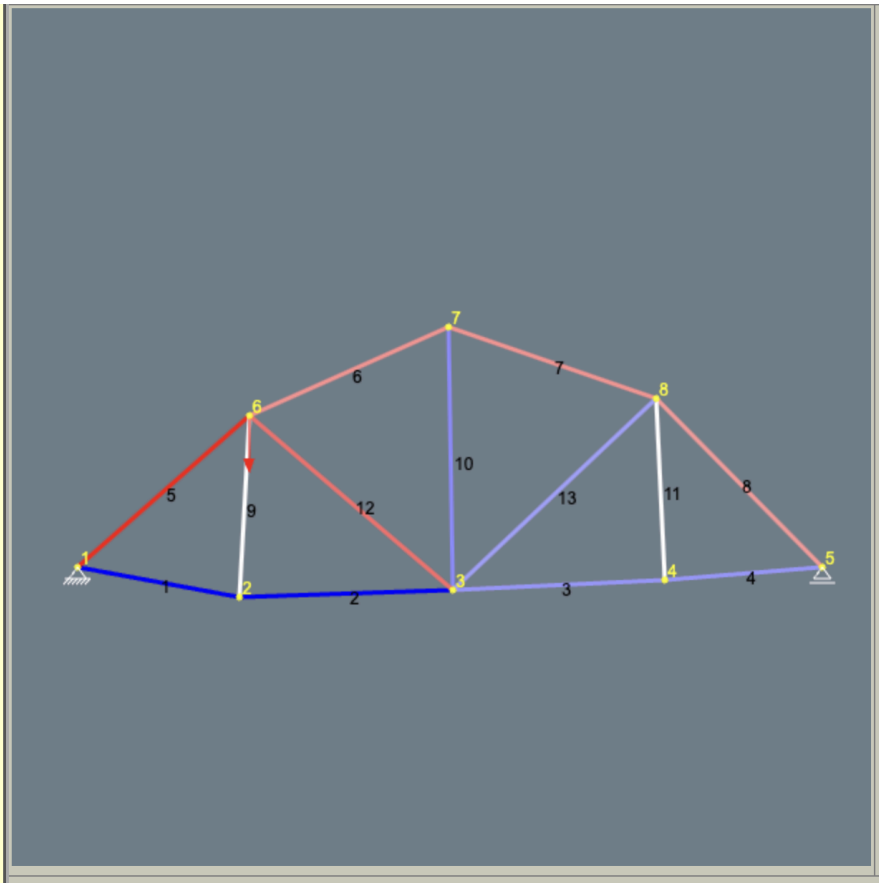
Method of Joints:



* Joint BC is meant to be 1.322 N

Part 4: Online Calculator

Loading Condition 1



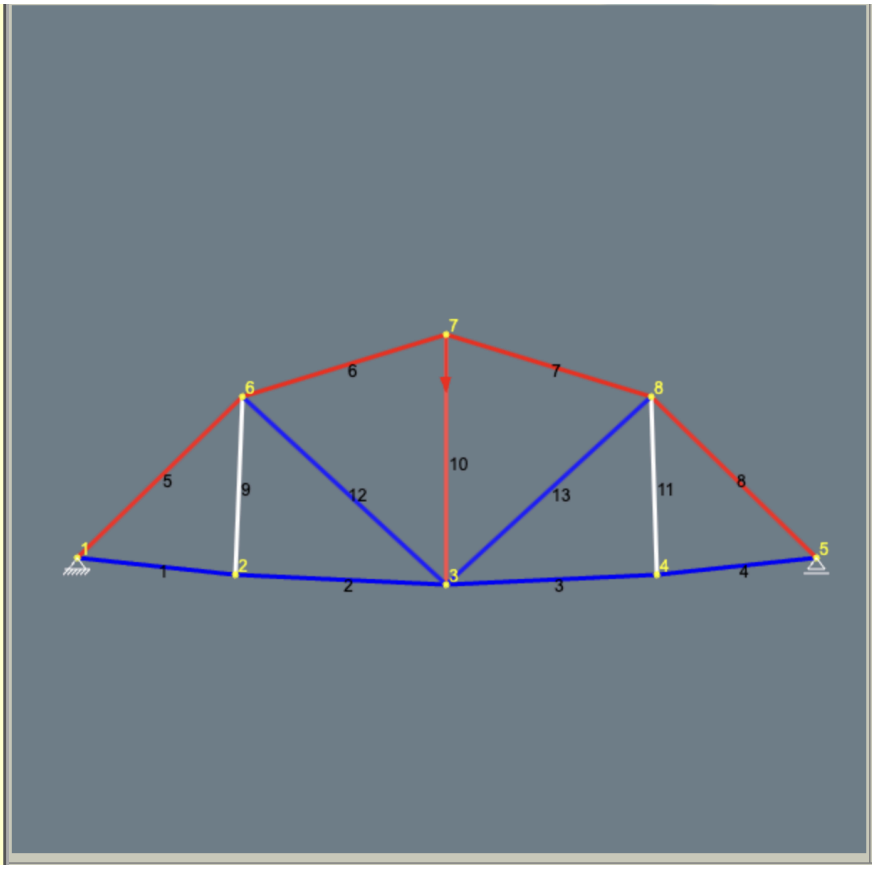
Truss Forces		
i	N_i	σ_i
1	1.321	5.285e-2
2	1.321	5.285e-2
3	0.3604	1.441e-2
4	0.3604	1.441e-2
5	-2.030	-8.121e-2
6	-0.6286	-2.515e-2
7	-0.6286	-2.515e-2
8	-0.5537	-2.215e-2
9	0.000	0.000
10	0.4415	1.766e-2
11	0.000	0.000
12	-0.9737	-3.895e-2
13	0.3033	1.213e-2

Reaction Forces		
i	F_{ix}	F_{iy}
1	0.000	1.542
5	0.000	0.4204

Displacements		
i	u_{ix}	u_{iy}
1	0.000	0.000
2	1.586e-4	-8.853e-4
3	3.700e-4	-6.712e-4

*Negative values/red coloring indicate the member is under compression, positive values/blue coloring indicate the member is under tension, and zero force members are in white

Loading Condition 2



Truss Forces		
i	N_i	σ_i
1	0.8409	3.363e-2
2	0.8409	3.363e-2
3	0.8409	3.363e-2
4	0.8409	3.363e-2
5	-1.292	-5.168e-2
6	-1.467	-5.867e-2
7	-1.467	-5.867e-2
8	-1.292	-5.168e-2
9	0.000	0.000
10	-0.9319	-3.728e-2
11	0.000	0.000
12	0.7076	2.831e-2
13	0.7076	2.831e-2

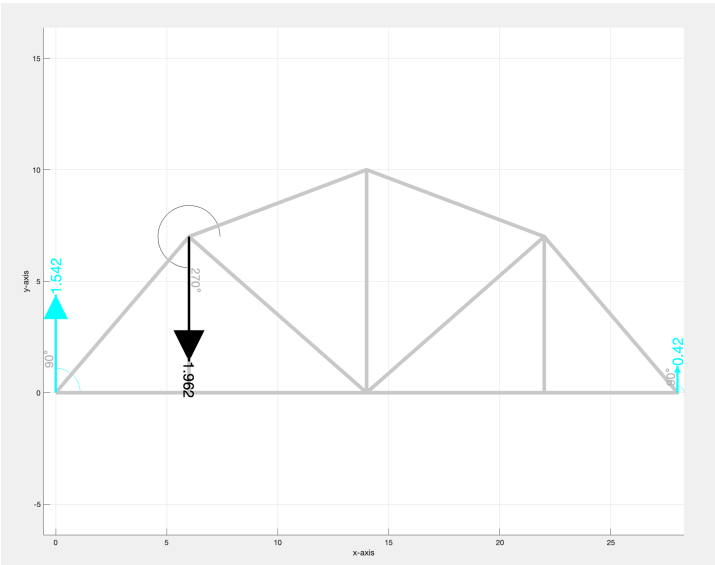
Reaction Forces		
i	F_{ix}	F_{iy}
1	0.000	0.9810
5	0.000	0.9810

Displacements		
i	u_{ix}	u_{iy}
1	0.000	0.000
2	1.009e-4	-5.829e-4
3	2.354e-4	-9.010e-4

Part 5: MATLAB Analysis

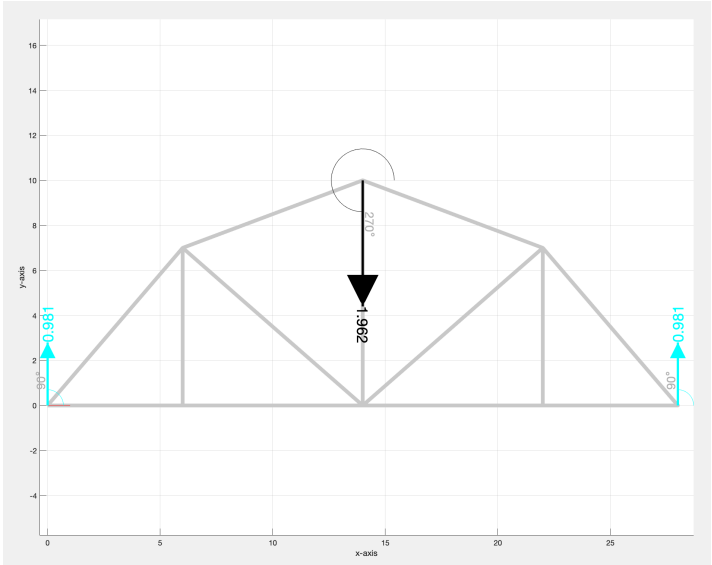
Loading Condition 1:

Reaction Forces of Entire Structure:



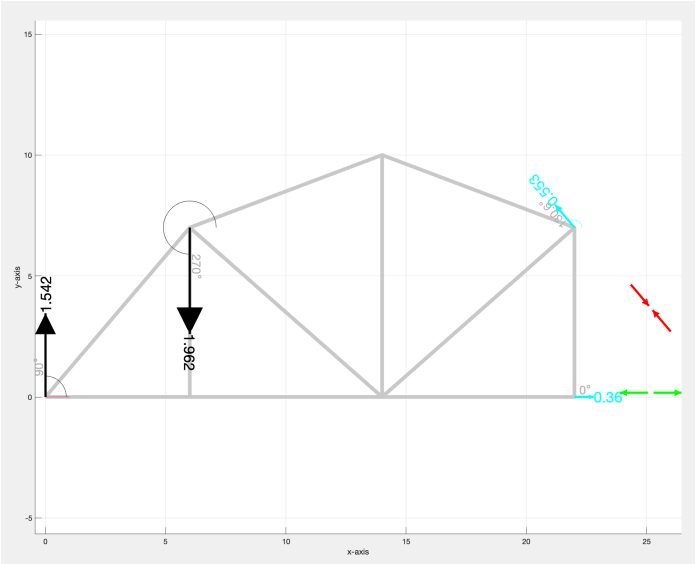
$A_y = 1.542\ N,\ E_y = 0.420\ N$

Loading Condition 2:

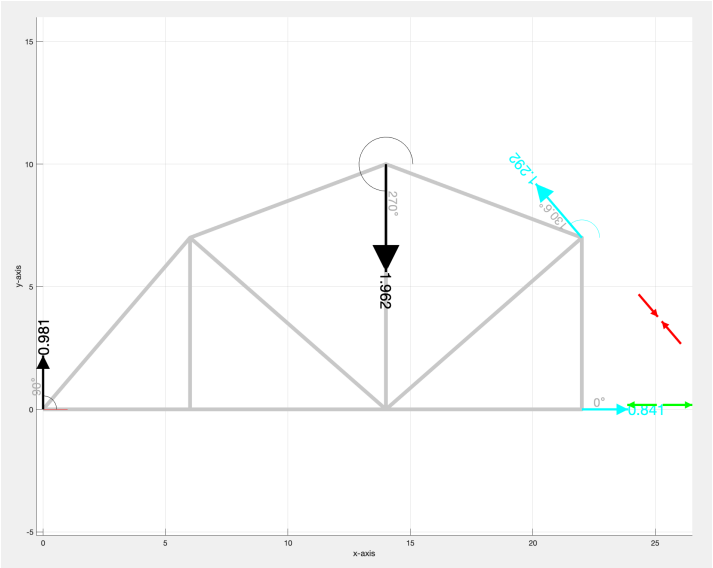


$A_y = 0.981\ N,\ E_y = 0.981\ N$

Internal Forces for Members DE and EH:

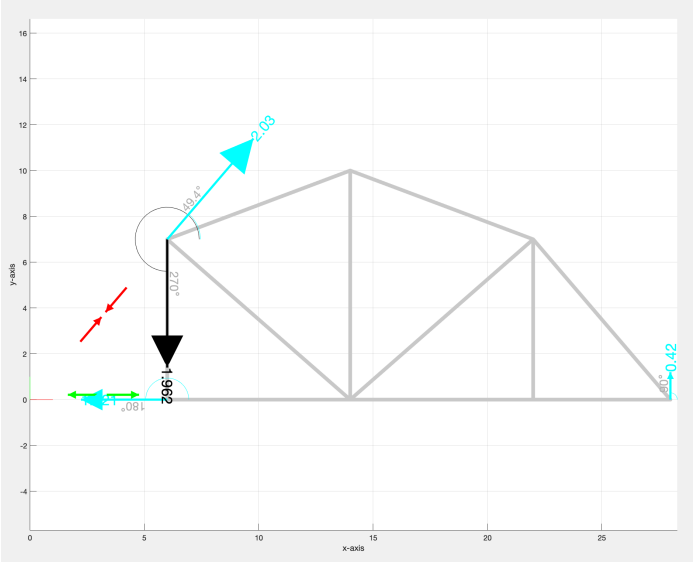


$F_{DE} = 0.360\ N,\ F_{EH} = 0.553\ N$

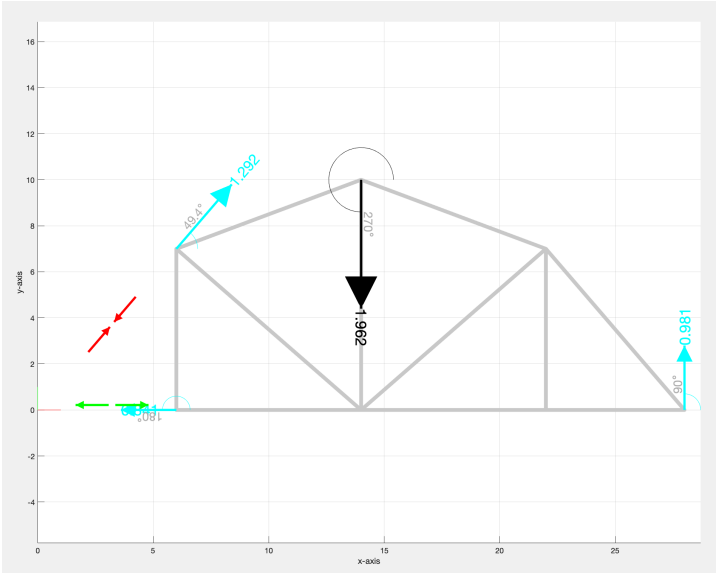


$F_{DE} = 0.841\ N,\ F_{EH} = 1.292\ N$

Internal Forces for Members AB and AF:

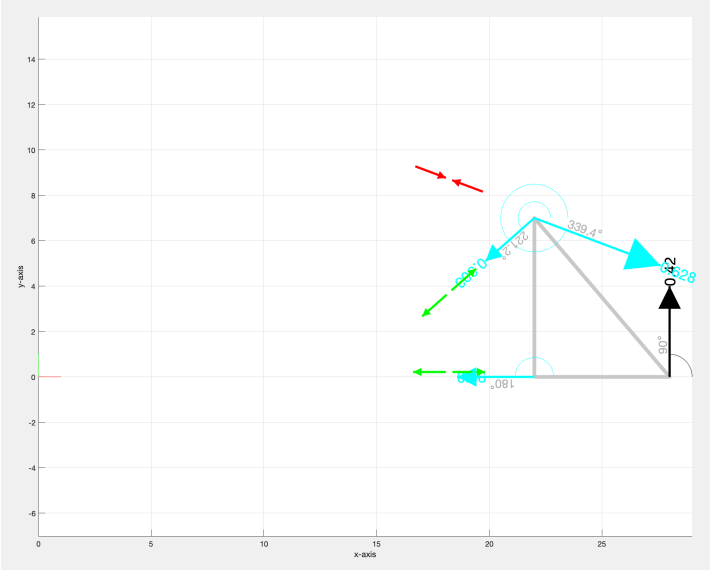


$F_{AB} = 1.321\ N,\ F_{AF} = 2.03\ N$

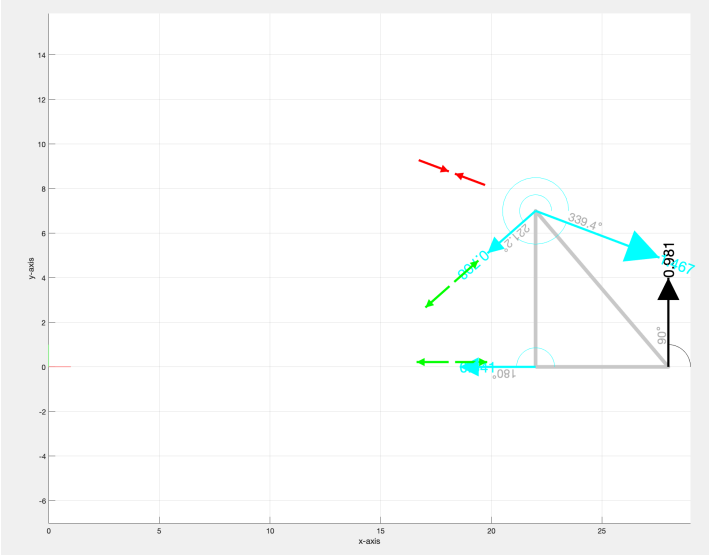


$F_{AB} = 0.841\ N,\ F_{AF} = 1.292\ N$

Internal Forces for Members CD, CH, and GH:

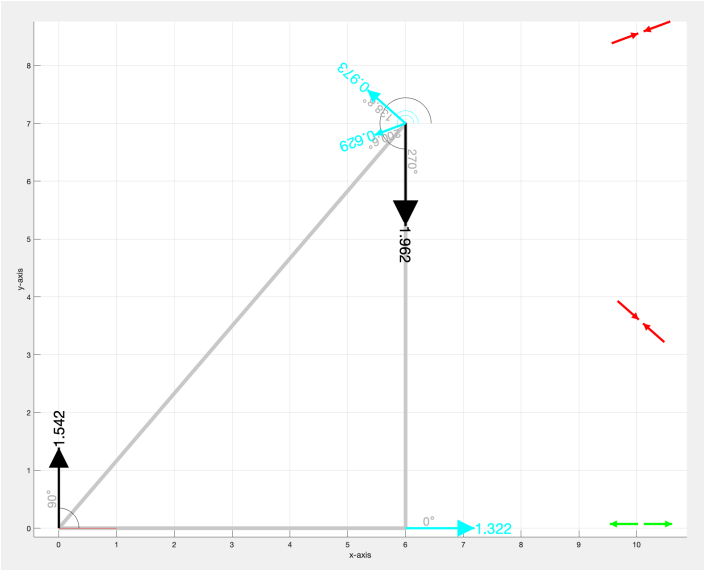


$F_{CD} = 0.360\ N,\ F_{CH} = 0.303\ N, F_{GH} = 0.628\ N$



$F_{CD} = 0.841\ N,\ F_{CH} = 0.708\ N, F_{GH} = 1.467\ N$

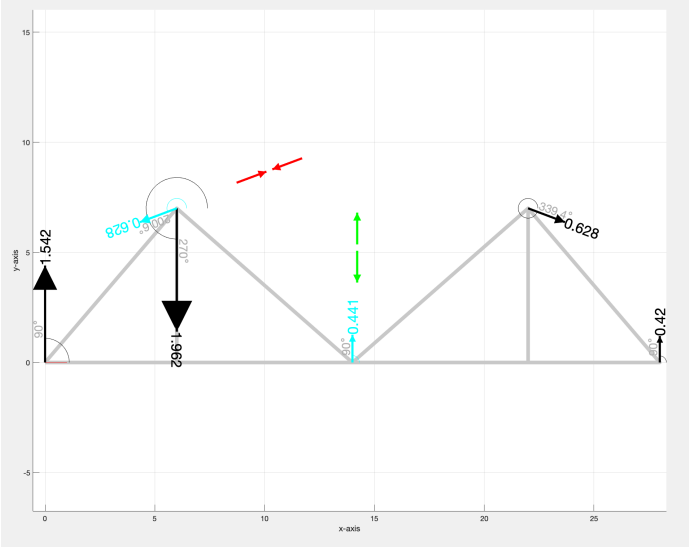
Internal Forces for Members BC, CF, and FG:



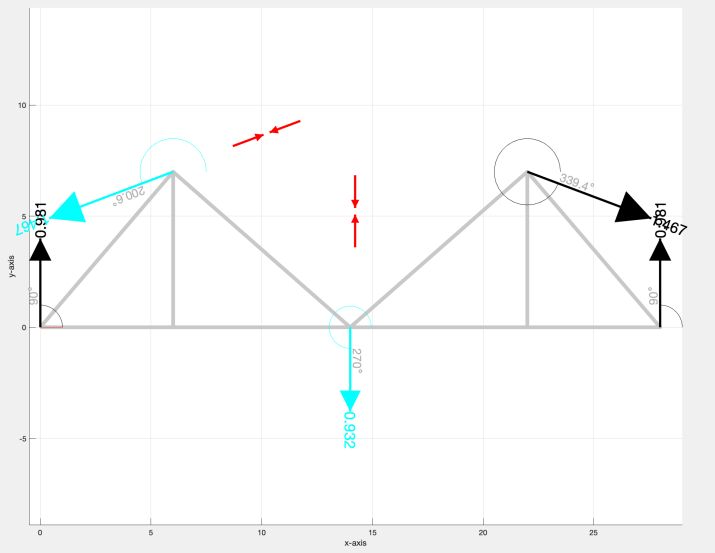
Can’t get it to work for condition 2 :(

$F_{BC} = 1.322\ N,\ F_{CF} = 0.973\ N, F_{FG} = 0.628\ N$

Internal Forces for Member CG:



$F_{CG} = 0.441\ N$



$F_{CG} = 0.932\ N,\ F_{FG} = 1.467\ N$

BF and DH are zero force members

Data Spreadsheet

Truss Data - Ani and Anne

https://docs.google.com/spreadsheets/d/1KB62OVoeyIq6dOhX34u6ExlphqEtyWCAvIMkFNMjHQI/edit?usp=sharing

Conclusion

The results from our theoretical calculations, the online truss calculator, and the MATLAB for the internal loads and reaction forces are all consistent with errors less than or equal to 0.2% for MATLAB and 0.8% for the online truss calculator. The physical loading and recorded masses indicated similar reaction forces to the theoretical calculations with an error less than or equal to 2%, which can likely be attributed to no centering the truss on the scales or to our truss not having exactly the same geometries physically as theoretically. We took as much care as possible while building the truss by hand to make the sides as symmetrical as possible, especially with the angles and lengths of the members. There were most likely small errors as we were measuring and cutting the pieces by hand, which would impact our results, however, our calculations show that our overall errors were very small, thus we consider our truss construction to be a success.

In this project, we were able to tangibly apply our knowledge of trusses and calculate internal loads on a real truss of our own design. During the design process, we learned what it means for a truss to be statically determinate, which is necessary for analyzing internal loads using the reaction forces and external loads. In developing our design, we also investigated various truss types and found that the Gambrel Truss is a type of roof truss commonly used in barns and sometimes homes and allows for a steeper pitch than traditional roofs. Through the internal load calculations, we were able to make the choice between using the method of joints and the method of sections. For some members, it was better to visualize and more efficiently solve multiple internal loads at a time using the method of sections. However, for others (like at joint F) method of joints was a much simpler option (especially since we were analyzing the loads of the entire truss rather than just a section). The online calculator was a great tool to check our calculations and help visualize what would happen to our truss under the loading conditions. We could see how a member being under tension or compression will actually affect the entire truss, and the similarities between the different loading conditions. We could discover for ourselves that the top members of bridges are usually in compression, while the bottom members are in tension, and understand why this makes sense. Beyond applying the knowledge we have gained in class about trusses, using MATLAB to analyze the internal loads and reaction forces allowed us to learn about the syntax of this useful coding language.

Overall, the various values we arrived at for the forces (internal and external) were consistent throughout our project with minimal error, and applying our knowledge about trusses from class was valuable in enhancing our understanding of these concepts.